# YIFEI SUN

Staff Software Engineer · Distributed Systems, Cloud Platform & API Infrastructure · 13 Years

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## SUMMARY

Platform engineer with 13 years building large-scale distributed services and cloud-native infrastructure. Deep expertise in Java (Concurrency, Collections, Streams, I/O), Python, REST/gRPC API design, and cloud infrastructure automation (AWS, Terraform, Kubernetes). Led cross-team redesign of a high-traffic platform API serving millions of daily requests (p99 250ms→50ms, availability 98%→99.9%). Recent work includes AI platform backends (MCP server architecture, LLM integration) and GenAI experimentation systems. Combines hands-on systems engineering with cross-organizational technical leadership across 10+ teams.

## LEADERSHIP HIGHLIGHTS

- Led cross-team platform API redesign: p99 5× reduction, 99.9% availability, Kubernetes-orchestrated microservices on AWS
- Built cloud infrastructure from scratch at early-stage startup — AWS provisioning, event pipelines, deployment automation
- Designed AI platform backends (MCP server, LLM integration) and GenAI experimentation systems from scratch
- Drove architecture standards and observability practices across 10+ teams; mentored engineers to Principal-level scope

**Technologies: Languages:** Java (Concurrency, Streams, I/O), Python, Node.js, Kotlin · **Infrastructure:** AWS (EC2, ECS, EKS, SQS, SNS, S3, RDS, DynamoDB), Terraform, Ansible, Kubernetes, Docker · **Data & Messaging:** Kafka, Redis, PostgreSQL, Cassandra, gRPC, GraphQL (Netflix DGS) · **Practices:** Microservices, Event-Driven, CQRS, Distributed Tracing, SLO Frameworks · Datadog, Grafana, OpenTelemetry · CI/CD

## INDEPENDENT PROJECT

### GenAI Decision Engine & Experimentation Platform

- Architected a hybrid rules + multi-path GenAI engine (fast vs. deep inference) with a policy overlay over heterogeneous telemetry, producing explainable real-time recommendations.
- Built "Prompt Lab" for end-to-end decision simulation, offline evaluation/backtesting, and prompt/parameter search (grid & Bayesian optimization) with multi-provider LLM support.
- Designed a configurable policy framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions.

## EXPERIENCE

### CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

#### High-Traffic Platform API Modernization

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core platform API (~8,000 rpm, millions of daily impressions), deploying as Kubernetes-orchestrated microservices on AWS.
- Introduced an orchestration service consolidating 12 downstream integrations into 5 well-defined services with per-service circuit breakers, timeout budgets, fault isolation, and idempotent retry.
- Built automated parity-testing infrastructure comparing new vs. legacy responses at 50% production traffic, catching dozens of edge-case bugs before they reached users.

  **p99: 250 ms → 50 ms** (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

- Defined architectural standards for new microservices; instrumented with structured logging, distributed tracing, and SLO-based latency budgets per downstream dependency.

#### AI Platform Backend — MCP Server & LLM Integration

- Sole backend engineer designing and implementing MCP server architecture for CarGurus' AI platform, exposing product capabilities to LLM-powered agents via tool-calling protocol.
- Delivered working POC in first sprint unblocking stakeholder demos; modular plugin architecture absorbed 3 rounds of requirement changes without rework. Expanded to designing APIs for multiple conversational AI agents across the org.

#### GraphQL Federation — Netflix DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into the federated GraphQL gateway, exposing data to frontend, backend, and AI consumers from a single graph.

- Implemented batch data loaders (N+1 prevention) and JVM warmup strategies eliminating cold-start latency spikes.

**Digital Commerce Platform — Event-Driven Architecture**

- Led backend design for a 0→1 commerce product on AWS using event-driven architecture (SQS/SNS, Kafka) with Terraform-managed infrastructure. Shipped under pandemic timeline pressure.
- Built infrastructure for rapid product experimentation — launch, measure, and safely retire product bets without destabilizing core systems.

**Cross-Org Platform Leadership & Standards**

- Established patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and architecture decision records.
- Mentored senior engineers on system design and path to Principal-level scope. Led 10+ technical training sessions.

### Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

**Authentication & Identity Platform**

- Upgraded legacy authentication to a modern Keycloak-based SSO platform on AWS, unifying auth across internal systems, enterprise partners, and external community (OAuth 2.0, SAML, social login). Zero-downtime migration via dual-auth bridge pattern.
- Built custom RBAC plugins for Kong API gateway (Python); deployed service mesh (Linkerd on Kubernetes) with mTLS for service-to-service trust.

**Event Sourcing & Microservice Architecture**

- Drove design and prototyping of microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.
- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations.

### Toast Inc. — Senior Software Engineer (early team, ~20 engineers)

Sep 2014 – Oct 2017

**Core Platform & Infrastructure (Early-Stage)**

- Among the earliest engineers during hypergrowth; designed and implemented core backend services (Order Service, Labor Service, REST APIs) that became the platform foundation for future growth.
- Built initial AWS environment (Ansible) — provisioned infrastructure, deployment pipelines, and monitoring before a dedicated DevOps team existed. Created first automated testing framework enabling safer releases.

**Third-Party Integrations & Partner Platform**

- Led integrations with key partners (workforce management, payment processors), ensuring reliable bidirectional sync. Partnered with PMs, sales, and support to translate customer feedback into platform capabilities.

### The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools (Java backend) for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

## EDUCATION

**M.S. in Electrical and Computer Engineering — Northeastern University, Boston** 2010 – 2013

**B.S. in Information Science — Beijing University of Aeronautics and Astronautics** 2006 – 2010